

STUDYING NEURONS USING THE HODGKIN-HUXLEY MODEL

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RESEARCH EXPERIENCE IN
COMPUTATIONAL SCIENCE
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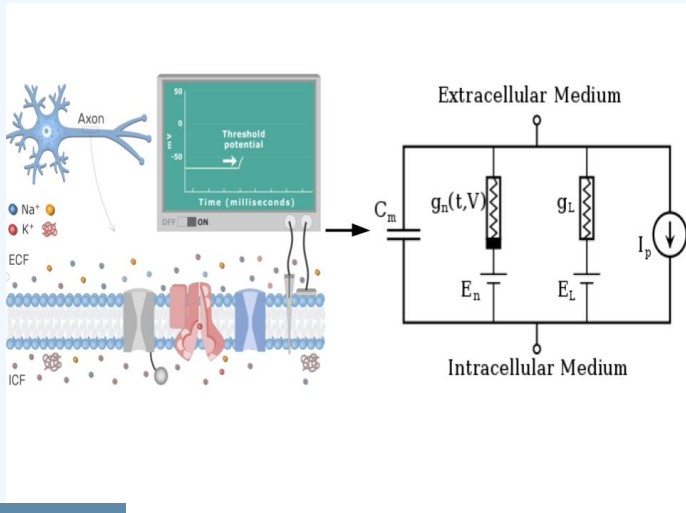
JUNE 28, 2024



NCSSM

RESEARCH QUESTION

How are membrane potentials affected by dynamic variables in the Hodgkin-Huxley model using NetLogo?



COMPUTATIONAL THINKING: TECHNOLOGIES

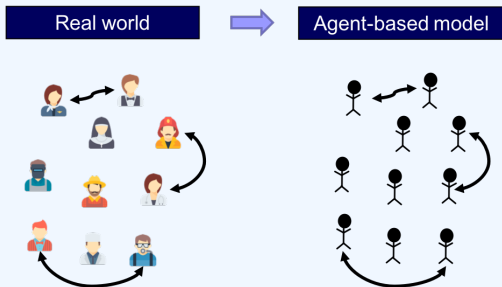
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COMPUTATIONAL THINKING: TECHNIQUES

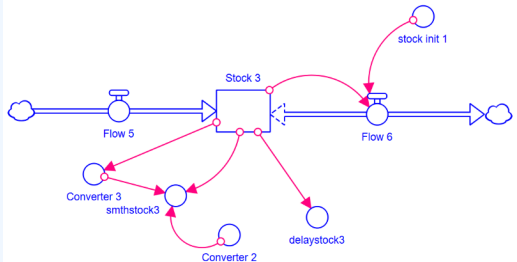
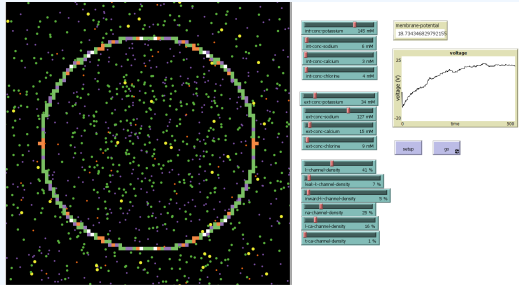
- Technique #1:
Differential Equations
- Technique #2:
Agent-Based Modeling

$$C_m \frac{dV_m}{dt} + I_{ion} = I_{ext}$$



COMPUTATIONAL THINKING: TOOLS

- Tool #1: Netlogo
 - ▶ Agent Simulation
- Tool #2: STELLA
 - ▶ Visualize Differential Equations



REFERENCES

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